

Melvin Abibula

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EDUCATION

Delft University of Technology — Bachelor's Degree in Computer Science and Engineering 09/2025 –
First Year Average Grade - 8.04/10
Relevant Modules - CSE Project (8.0), Web & Database Technologies (8.0), Algorithms & Data Structures (8.5),
Software Quality & Testing (8.5)
Liceul Teoretic Internațional de Informatica - Constanta, Romania 09/2021 – 07/2025
• *Romanian Baccalaureate Diploma* – Mathematics and Computer Science Specialisation – 9.46/10 Average

ACADEMIC & TECHNICAL EXPERIENCE

National Olympiad in Informatics - Bucharest, Romania 04/2024

- Awarded **Silver Medal and Special Mention (MEN)** at the national level
- Solved advanced algorithmic problems involving dynamic programming, graph theory, and combinatorics
- Implemented optimized C++ solutions under strict time and memory constraints
- Ranked among the top competitors nationwide

LEADERSHIP AND EXTRACURRICULAR

Game Development Club – Founder & President

Liceul Teoretic Internațional de Informatica 2023-2024

- Founded and led a 10+ member club focused on game development and programming
- Developed leadership, mentoring, and project management skills by organizing interactive workshops and guiding members through technical projects
- Mentored peers in designing and implementing their first 2D games using Unity and C#
- Planned and coordinated weekly sessions, delegated tasks, and ensured milestone completion, improving team productivity and engagement

Chess Club – President

Liceul Teoretic Internațional de Informatica 2023-2024

- Established and managed a 15+ member chess club, promoting analytical thinking and strategic planning
- Developed leadership and coaching abilities by organizing structured training sessions and internal tournaments

PROJECTS

Custom NES Emulator

C++

- Developed a custom Nintendo Entertainment System (NES) emulator, implementing core hardware components and low-level system architecture
- Programmed the NES 6502 CPU core logic, translating opcodes and managing complex memory addressing modes
- Engineered Picture Processing Unit (PPU) bus registers to handle system memory synchronization and data state
- Implemented the Mapper 000 (NROM) protocol to route static PRG and CHR cartridge memory directly into the CPU and PPU address spaces
- Validated emulation accuracy by executing testing routines against the canonical nestest ROM and booting games like *Donkey Kong*

FoodPal — Distributed Cooking Organizer Application

CSE Project, Delft University of Technology
Java, Spring Boot, JavaFX, JSON, Jackson

- Collaborated in a team of 4 to develop a distributed client-server cooking organizer application for recipe management and meal planning
- Implemented core recipe management features including creating, editing, deleting, cloning, and browsing recipes in a distributed client-server architecture

- Developed the full-text search system, enabling filtering by recipe names, ingredients, and preparation steps with support for multi-term queries and recipe sorting
- Built the live language switch feature using JavaFX internationalization, supporting dynamic runtime translation, persistent language preferences, and recipe language filtering
- Extended localization support by adding Romanian, Bulgarian, and German alongside English and Dutch

SKILLS AND INTERESTS

Technical Skills:

- **Programming Languages:** C++, C#, Java, JavaScript, SQL
- **Frameworks & Libraries:** Spring Boot, JavaFX
- **Tools:** Git
- **Web Technologies:** HTML, CSS
- **Other Technical Skills:** Competitive Programming, Data Structures & Algorithms, Problem Solving

Languages: English (C1), Romanian (Native)

Certifications: International Computer Driving Licence